

Practice Problems (Branching)

Required knowledge

Basic C programming, Relational operators, Logical operators

Level 1: Simple `if` Statement

1. **Check Positive Number:** Take an integer as input. Use an `if` statement to print "Positive" if the number is greater than zero.
2. **Check Even or Odd:** Prompt the user to enter an integer. Use an `if` statement to print "Even" if the number is divisible by 2.
3. **Check Vowel:** Take a single character as input and use an `if` statement to check if it's a vowel (a/A, e/E, i/I, o/O, u/U).
4. **Eligible for Voting:** Take the user's age as input and print "Eligible to vote" if they are 18 or older.
5. **Check Uppercase Letter:** Take a character as input and print "Uppercase" if the character is an uppercase letter.

Level 2: `if...else` Statement

6. **Number Sign Checker:** Take an integer as input and print "Positive" if it's greater than zero, otherwise print "Non-positive."
7. **Temperature Check:** Prompt the user to enter a temperature in Celsius and print "Above freezing" if the temperature is above 0°C, otherwise print "At or below freezing."
8. **Pass/Fail:** Take a student's score as input. If the score is 50 or more, print "Pass"; otherwise, print "Fail."
9. **Check Divisibility by 5:** Take an integer as input and print "Divisible by 5" if the number is divisible by 5, otherwise print "Not divisible by 5."
10. **Leap Year Checker:** Take a year as input and print "Leap year" if it is divisible by 4 and not divisible by 100, or if it's divisible by 400. Otherwise, print "Not a leap year."

Level 3: `if...else if...else` Ladder Statement

11. **Grade Calculator:** Take a student's score as input and print the grade based on the score:
 - 90 or above: "A"
 - 80–89: "B"
 - 70–79: "C"
 - 60–69: "D"
 - Below 60: "F"
12. **Number Range:** Take an integer as input and print if it's "Small" (1–10), "Medium" (11–100), or "Large" (101 and above).
13. **Number Comparison:** Take two integers as input and print whether the first number is "Greater than," "Less than," or "Equal to" the second number.
14. **Temperature Feedback:** Take a temperature as input and print:
 - "Cold" if below 10°C
 - "Warm" if between 10°C and 25°C
 - "Hot" if above 25°C
15. **Body Mass Index (BMI) Category:** Take a user's weight (kg) and height (m), calculate the BMI, and categorize it as:
 - Underweight (< 18.5)
 - Normal weight (18.5–24.9)
 - Overweight (25–29.9)
 - Obese (30 and above)

Level 4: Nested `if...else` Statements

16. **Positive/Negative Even or Odd:** Take an integer as input and check if it is:
 - Positive even
 - Positive odd
 - Negative even
 - Negative odd
17. **Triangle Type Checker:** Take three side lengths as input and check if they form:
 - An equilateral triangle (all sides equal)
 - An isosceles triangle (two sides equal)
 - A scalene triangle (all sides different)
18. **Admission Eligibility:** Prompt for a student's marks in Math and Science. To be eligible for admission:

- Math and Science both must be 70 or more.
 - If both are above 90, print "Eligible for scholarship."
- 19. Nested Divisibility Check:** Take an integer as input, check if it is:
- Divisible by both 5 and 10
 - Divisible only by 5
 - Divisible only by 10
- 20. Student Honors and Distinction:** Take a student's score as input:
- If the score is 85 or above, print "Distinction."
 - If it's between 70 and 84, print "First Class."
 - If between 50 and 69, print "Second Class."
 - If below 50, print "Fail."

Level 5: `switch` Statement

- 21. Day of the Week:** Take a number (1–7) as input and print the corresponding day of the week using a `switch` statement.
- 22. Month Days:** Take a number (1–12) and print the number of days in that month using `switch` (assume February has 28 days).
- 23. Calculator Using Switch:** Take two numbers and an operator (+, -, *, /) as input and perform the respective operation using a `switch` statement.
- 24. Season Finder:** Take a month number (1–12) as input and print the season (e.g., "Winter," "Spring," etc.) using a `switch` statement.
- 25. Grading System with Switch:** Take a grade (A, B, C, D, F) as input and print the corresponding description using `switch`:
- A: "Excellent"
 - B: "Good"
 - C: "Average"
 - D: "Below Average"
 - F: "Fail"

Level 6: Conditional/Ternary Operator (?:)

- 26. Absolute Value:** Take an integer as input and use the ternary operator to print its absolute value.
- 27. Even or Odd (Ternary):** Take an integer and use the ternary operator to print "Even" if it's even and "Odd" if it's odd.
- 28. Larger of Two Numbers:** Take two integers and use the ternary operator to print the larger one.

29. **Minimum of Three Numbers:** Take three integers and use nested ternary operators to print the smallest of the three.
30. **Grade Status:** Take a student's score as input and use the ternary operator to print "Pass" if it's 50 or above, and "Fail" otherwise.

Level 7: Real-World Applications with Mixed Branching

31. **Shopping Discount Calculator:** Take a purchase amount as input:
- If it's above \$1000, apply a 10% discount.
 - Otherwise, print the total amount without discount.
32. **Age Category:** Take the user's age as input and print their age category:
- Child (0–12)
 - Teen (13–19)
 - Adult (20–59)
 - Senior (60 and above)
33. **Electricity Bill Calculator:** Take units consumed as input and calculate the bill as follows:
- First 100 units: \$0.50 per unit
 - Next 100 units: \$0.75 per unit
 - Above 200 units: \$1.20 per unit
34. **Simple Tax Calculator:** Take an income amount as input and calculate tax as follows:
- Income up to \$10,000: no tax
 - Income between \$10,001 and \$20,000: 10% tax
 - Income above \$20,000: 20% tax
35. **Exam Result Summary:** Take marks in three subjects as input:
- If all marks are 50 or above, print "Pass."
 - If any mark is below 50, print "Fail."